AWS VPC Peering

Connect two VPCs. Prove private connectivity.

Student project report: EC2, custom route tables, security groups and a bastion host

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Batch	1
Project Title	AWS VPC Peering
Region	Asia Pacific (Mumbai)/ap-south-1
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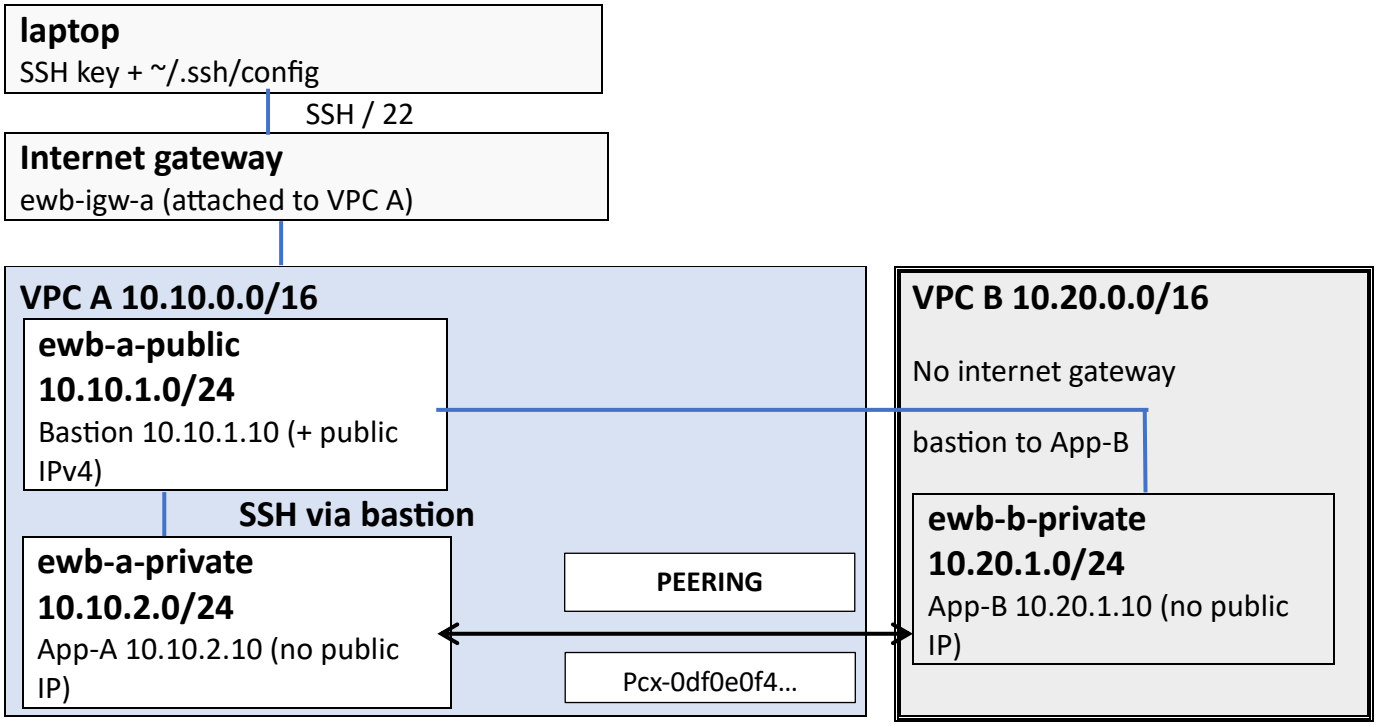
1. Project overview

EWB has an application in VPC A and a second workload in VPC B. Both application servers must stay private. The goal was to connect the two VPCs with a VPC peering connection so the servers can exchange traffic over their private IPv4 addresses, with one public bastion host in VPC A used for administration.

What was demonstrated

1) Non-overlapping VPC and subnet CIDRs.
2) Public and private subnets with explicit route table associations.
3) Peering activated, return routes configured and only the required inbound traffic allowed.

2. Architecture



Administration path: laptop to bastion to App-A or App-B using SSH Proxy Jump. Project traffic: App-A and App-B talk over private IPs through the peering connection. VPC B has no internet gateway and peering does not give VPC B internet access through VPC A.

3. Resource inventory

Resource	Name	ID/address
VPC A	ewb-A	vpc-023221eb47de6c02b / 10.10.0.0/16
VPC B	ewb-B	vpc-0737d8a77facdc977 / 10.20.0.0/16
Subnet	ewb-a-public	subnet-075016e5fde052adc / 10.10.1.0/24 / ap-south-1a
Subnet	ewb-a-private	subnet-0957ac630c43a018a / 10.10.2.0/24 / ap-south-1a
Subnet	ewb-b-private	subnet-0592a8d7ef4b1d8d0 / 10.20.1.0/24 / ap-south-1b
Internet gateway	ewb-igw-a	igw-029d1f89c2e2458c3 (attached to VPC A)
Route table	ewb-rt-a-public	rtb-097d2c15000e4068a
Route table	ewb-rt-a-private	rtb-0ec917b9fd110d2cc
Route table	ewb-rt-b-private	rtb-0f010b4bd768d9c0a
Security group	ewb-sg-bastion	sg-07b4031abd15552f5 (VPC A)
Security group	ewb-sg-app-a	sg-0ce76db0d81273b2a (VPC A)
Security group	ewb-sg-app-b	sg-065323c11103bfdfd (VPC B)
Peering connection	ewb-a-b-peer	pcx-0df0e0f4805a7d042 (Active)
EC2 instance	ewb-bastion	i-0adc6e7ebd00a6364 / 10.10.1.10 / public 65
EC2 instance	ewb-app-a	i-03d0e8cb5bfbf5de7 / 10.10.2.10 / no public

EC2 instance	ewb-app-b	i-0c47fb24ae34ae58c / 10.20.1.10 / no public IP / ap-south-1b
Key pair	ewb-lab	RSA, .pem (private key not included in this r

4. Configuration

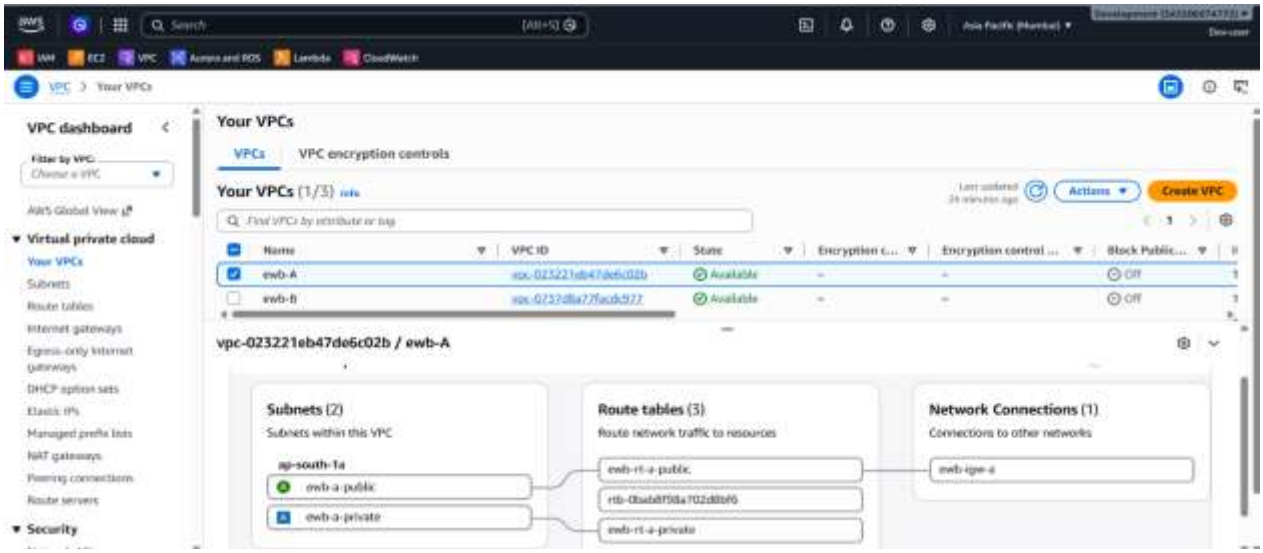
4.1 Route tables (final state)

Route table	Associated subnet	Destination	Target	Purpose
ewb-rt-a-public	ewb-a-public	10.10.0.0/16	local	Bastion to App-A
		0.0.0.0/0	ewb-igw-a	Bastion internet access
		10.20.0.0/16	pcx-0df0e0f4805a7d042	Bastion SSH to App-B (ProxyJump)
ewb-rt-a-private	ewb-a-private	10.10.0.0/16	local	Local VPC A
		10.20.0.0/16	pcx-0df0e0f4805a7d042	App-A to VPC B
ewb-rt-b-private	ewb-b-private	10.20.0.0/16	local	Local VPC B
		10.10.0.0/16	pcx-0df0e0f4805a7d042	Return path to VPC A

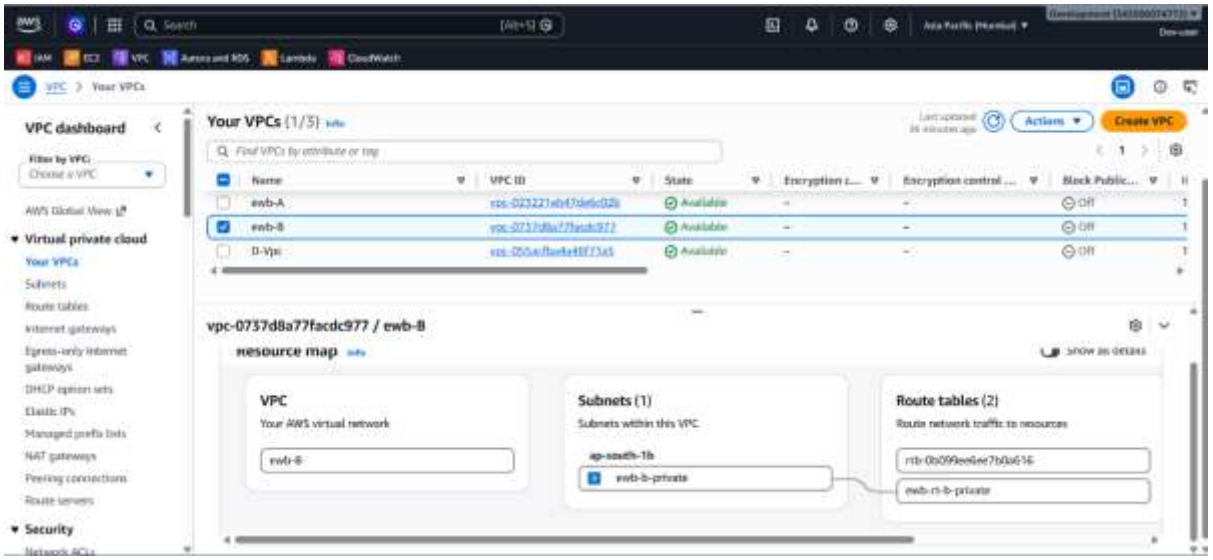
4.2 Security group inbound rules

Security group	Type / port	Source	Reason
ewb-sg-bastion	SSH / TCP 22	My public IPv4 /32	Only my laptop can reach the bastion
ewb-sg-app-a	SSH / TCP 22	10.10.1.10/32	Admin only through the bastion
	ICMP Echo Request	10.20.1.10/32	Ping from App-B
ewb-sg-app-b	SSH / TCP 22	10.10.1.10/32	Proxy Jump arrives from the bastion private IP
	ICMP Echo Request	10.10.2.10/32	Ping from App-A

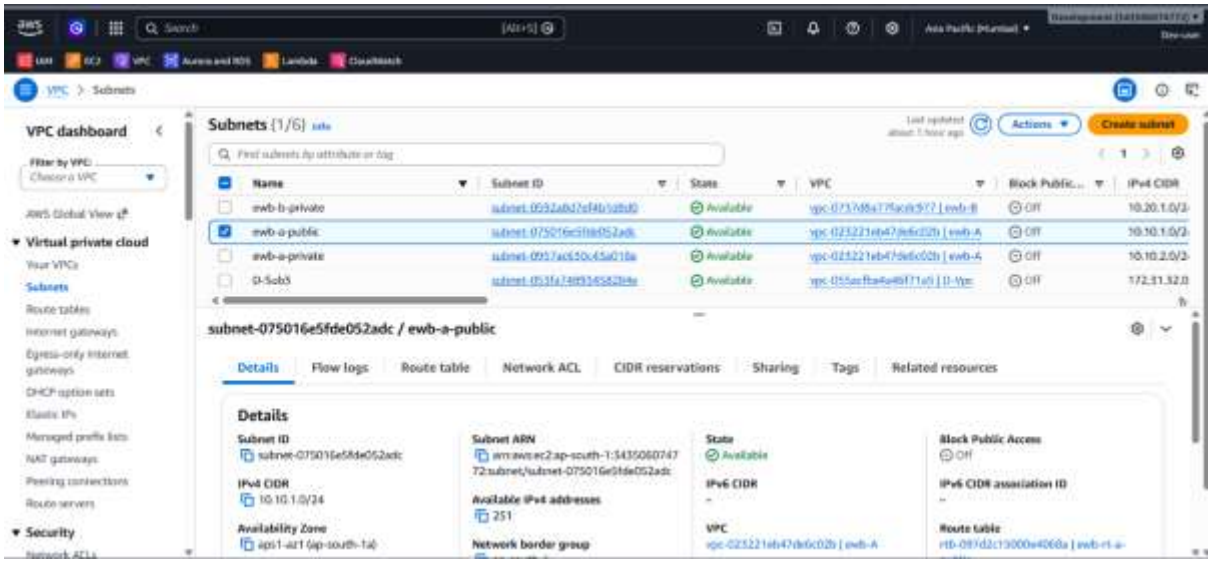
5. Evidence: network build



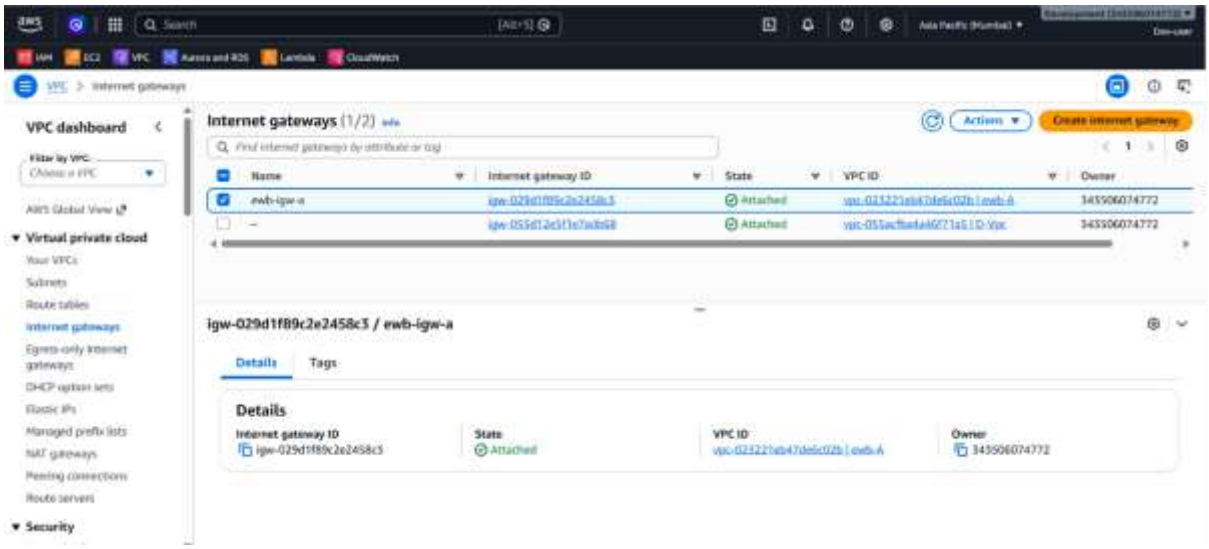
Checkpoint 01 - VPC A resource map: ewb-a-public uses ewb-rt-a-public (to ewb-igw-a); ewb-a-private uses ewb-rt-a-private with no internet route.



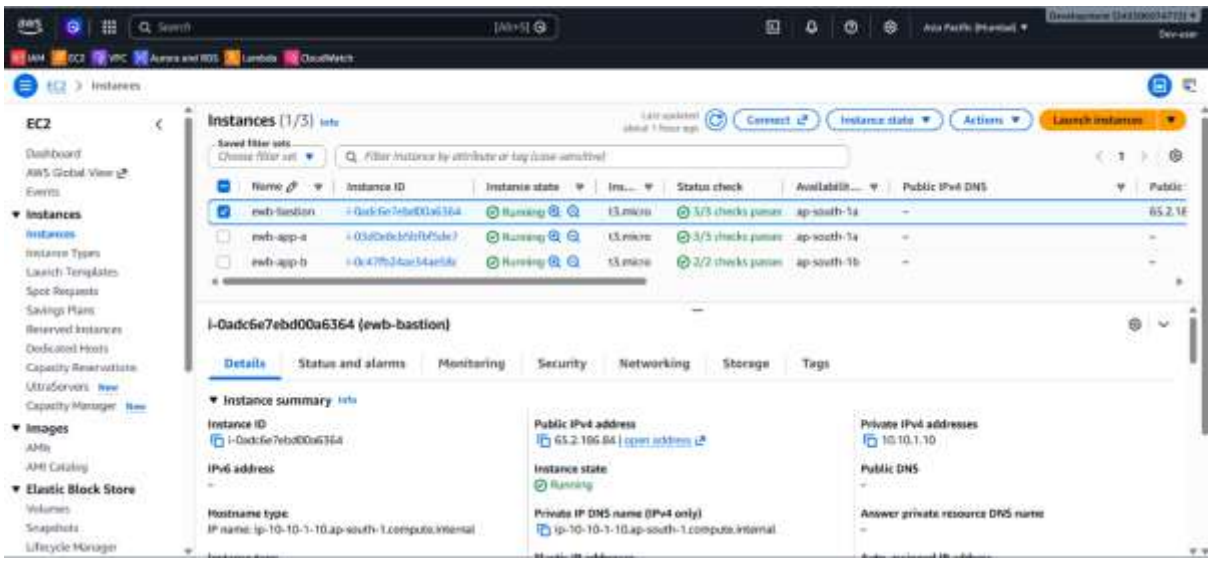
VPC B resource map: ewb-b-private uses ewb-rt-b-private; VPC B has no internet gateway.



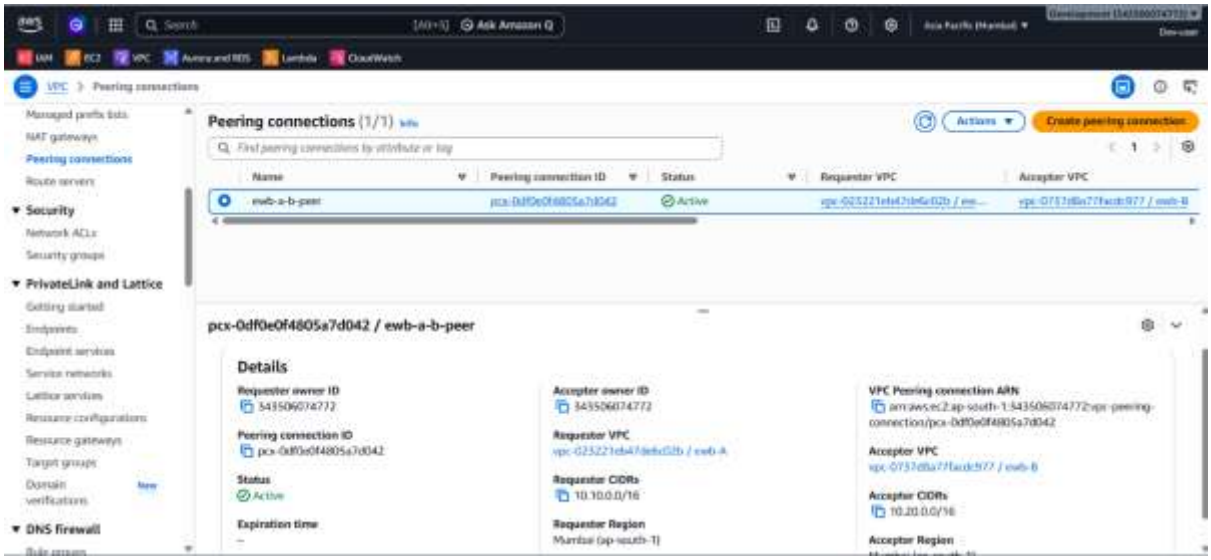
Subnets: ewb-a-public (10.10.1.0/24), ewb-a-private (10.10.2.0/24), ewb-b-private (10.20.1.0/24).



Internet gateway ewb-igw-a attached to vpc-023221eb47de6c02b (ewb-A)

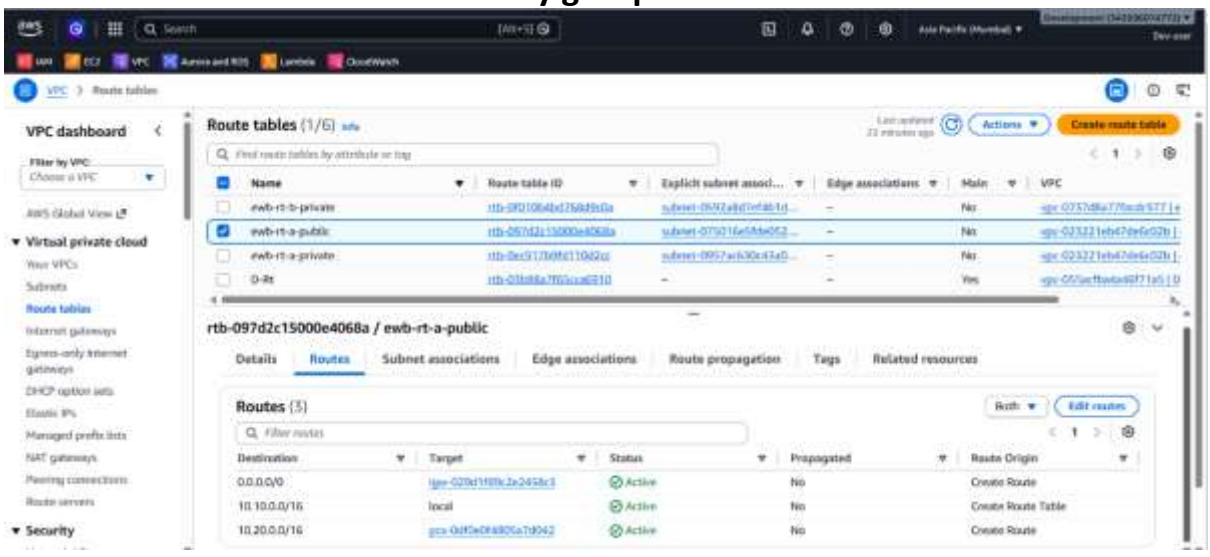


Checkpoint 02 - three instances running, status checks passed; only ewb-bastion has a public IPv4 (65.2.186.84), private IP 10.10.1.10.

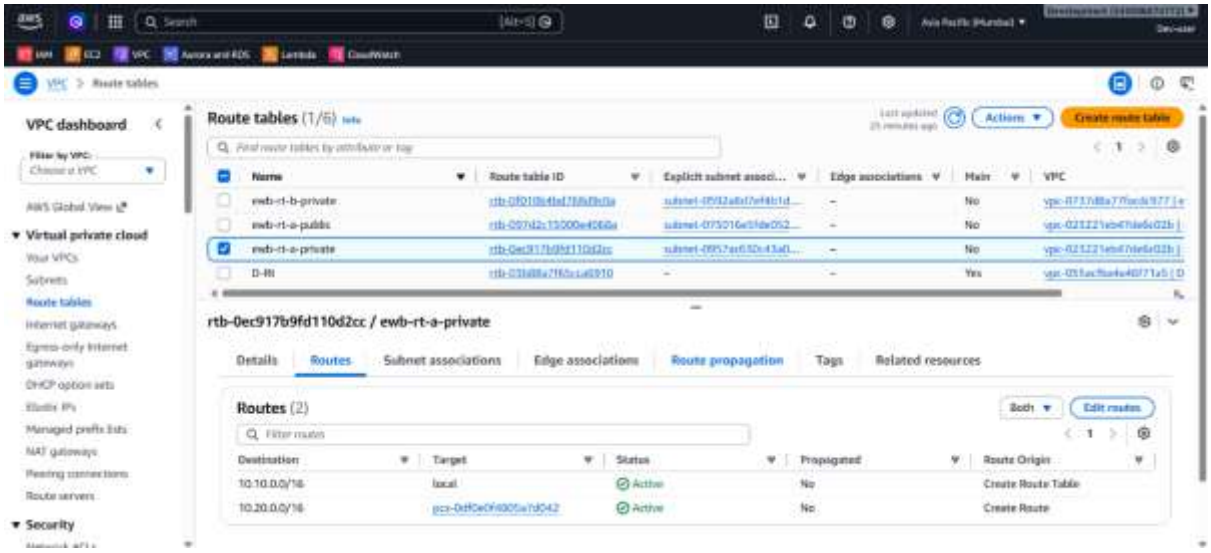


Checkpoint 03 - peering connection ewb-a-b-peer is Active: requester vpc-023221eb47de6c02b (10.10.0.0/16), accepter vpc-0737d8a77facdc977 (10.20.0.0/16

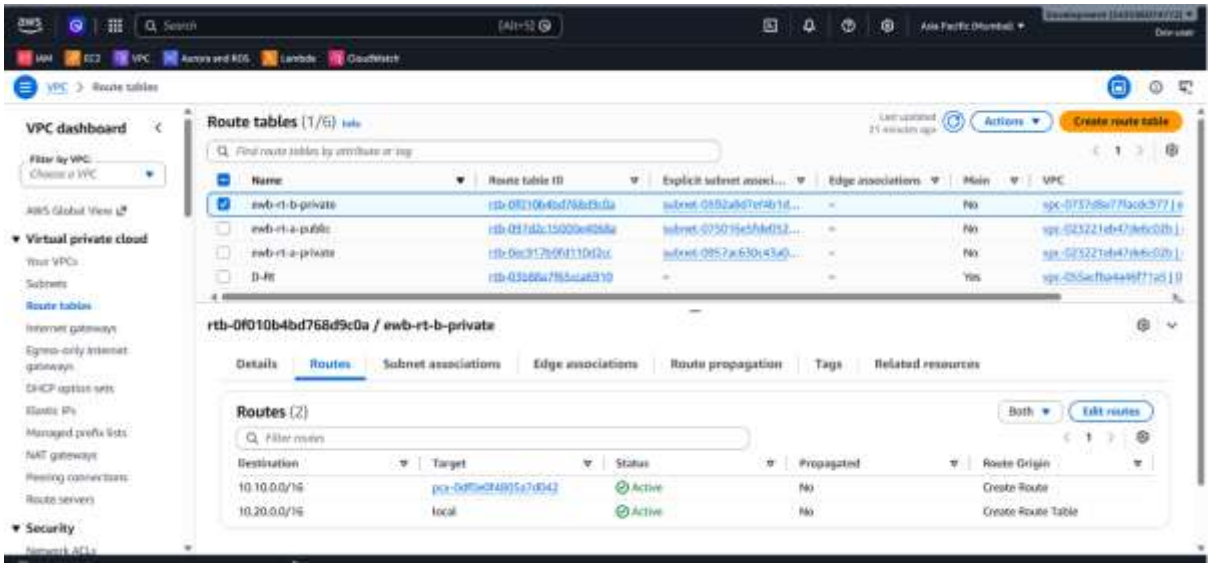
6. Evidence: routes and security groups



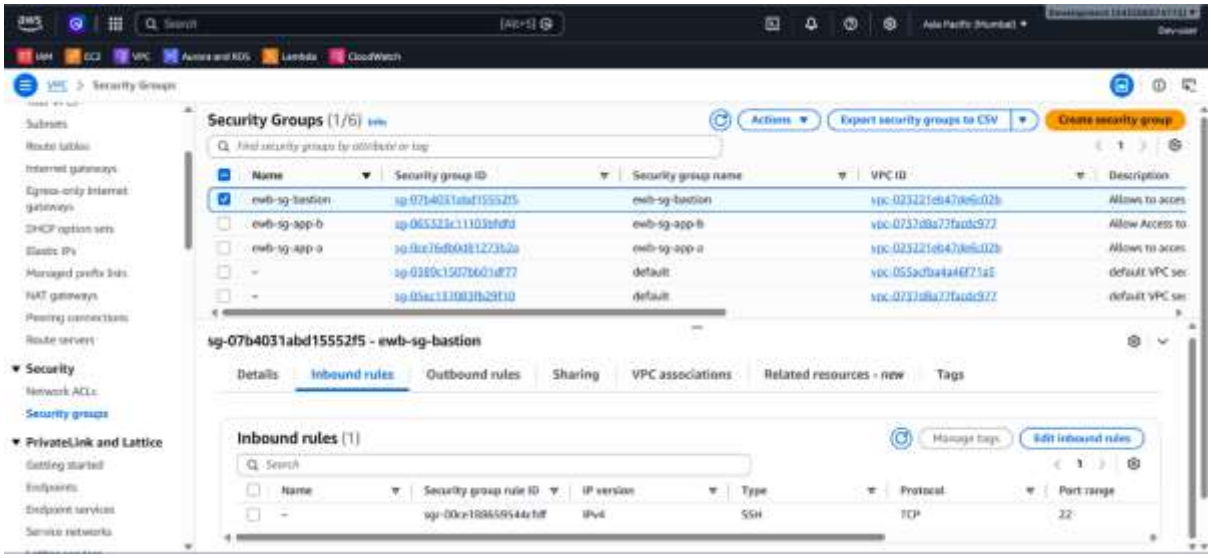
Checkpoint 04 - ewb-rt-a-public: local, 0.0.0.0/0 to the internet gateway, 10.20.0.0/16 to the peering connection.



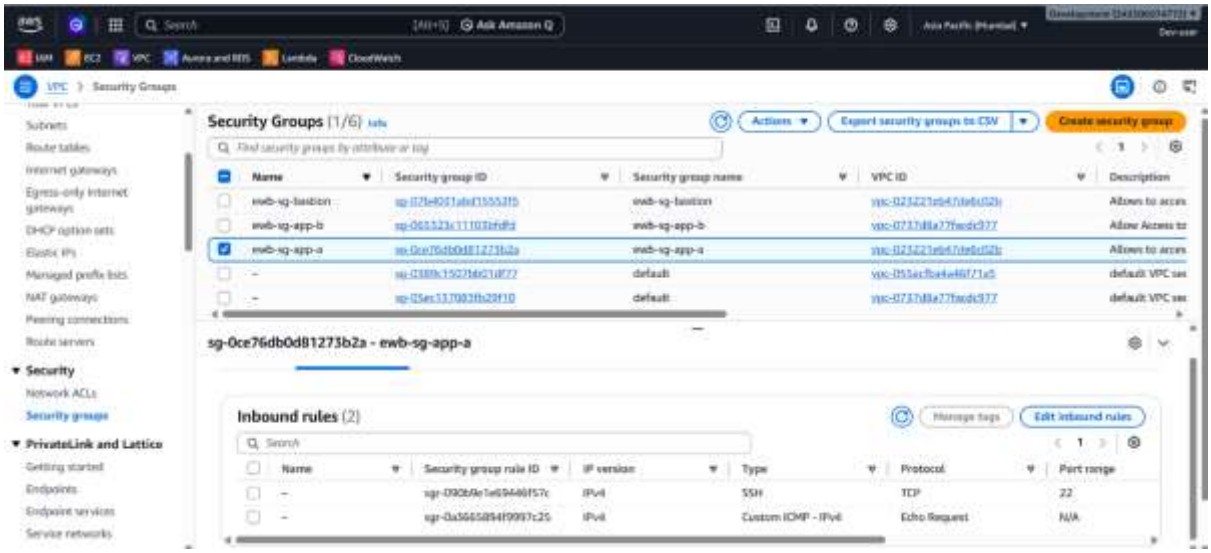
ewb-rt-a-private: local and 10.20.0.0/16 to the peering connection. No internet route



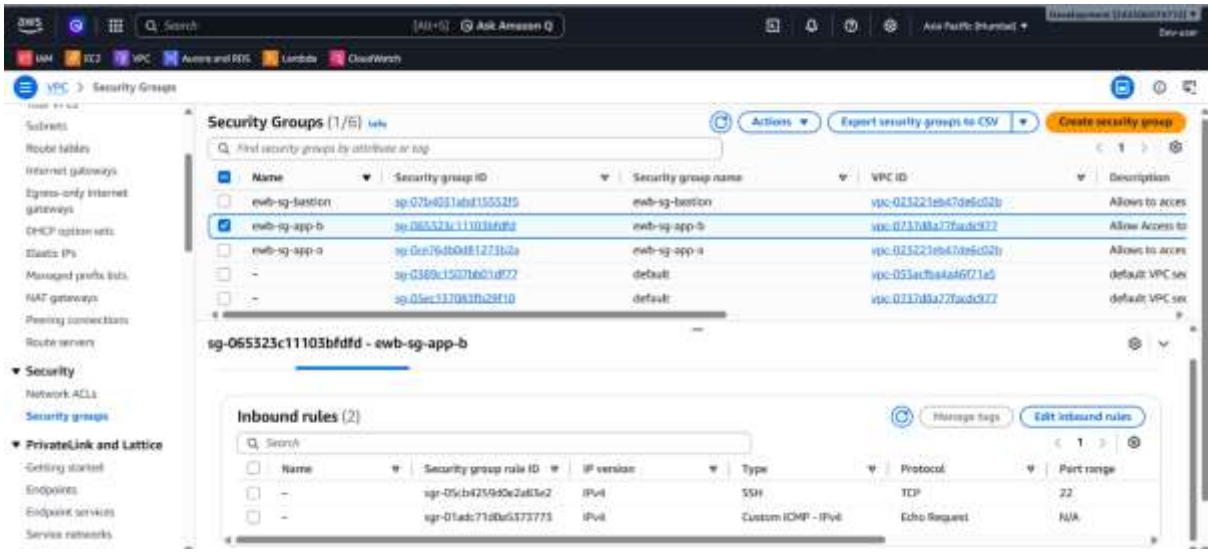
ewb-rt-b-private: local and 10.10.0.0/16 to the peering connection (return route).



ewb-sg-bastion inbound: one rule, SSH TCP 22 from my /32 address.



ewb-sg-app-a inbound: SSH TCP 22 and ICMP Echo Request (2 rules).

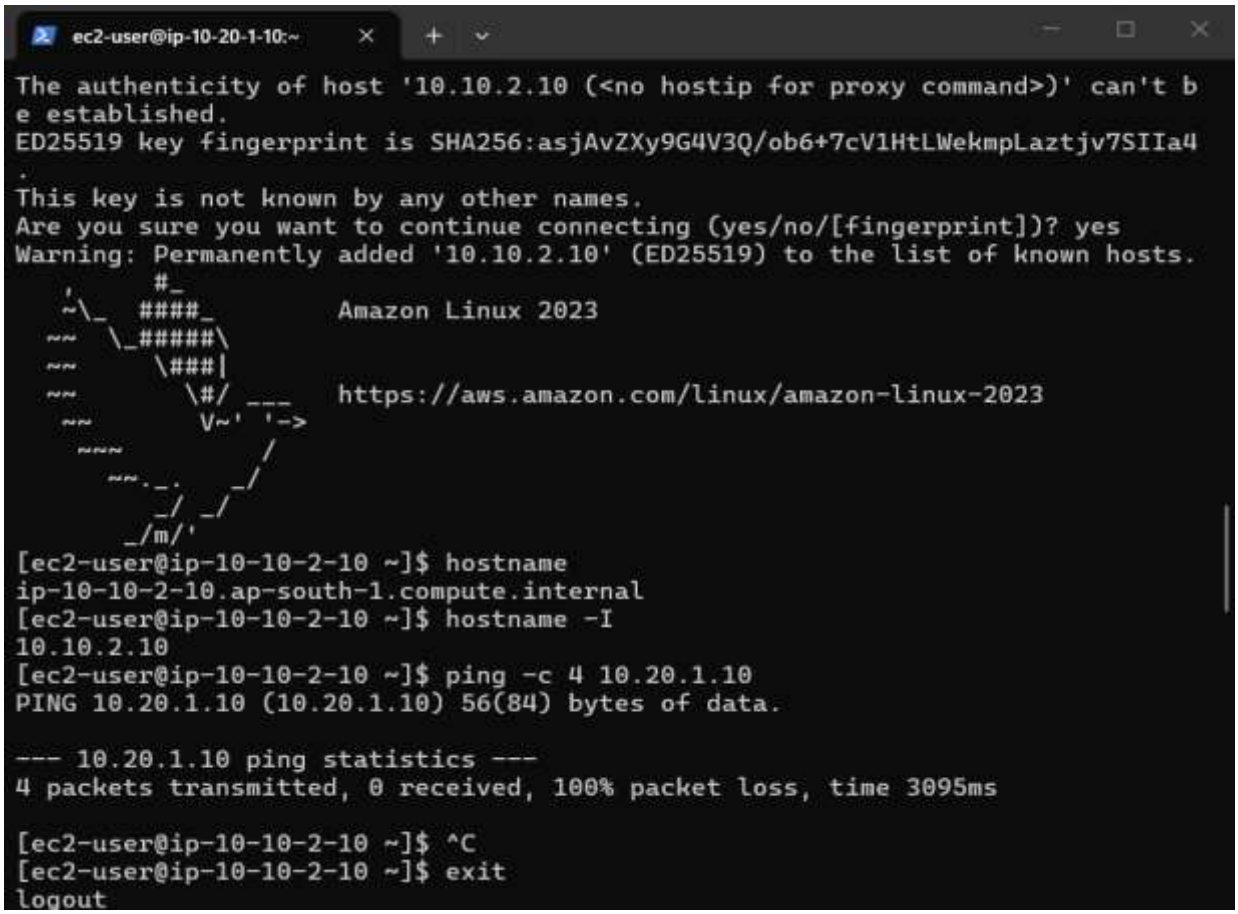


ewb-sg-app-b inbound: SSH TCP 22 and ICMP Echo Request (2 rules, after the ICMP rule was restored).

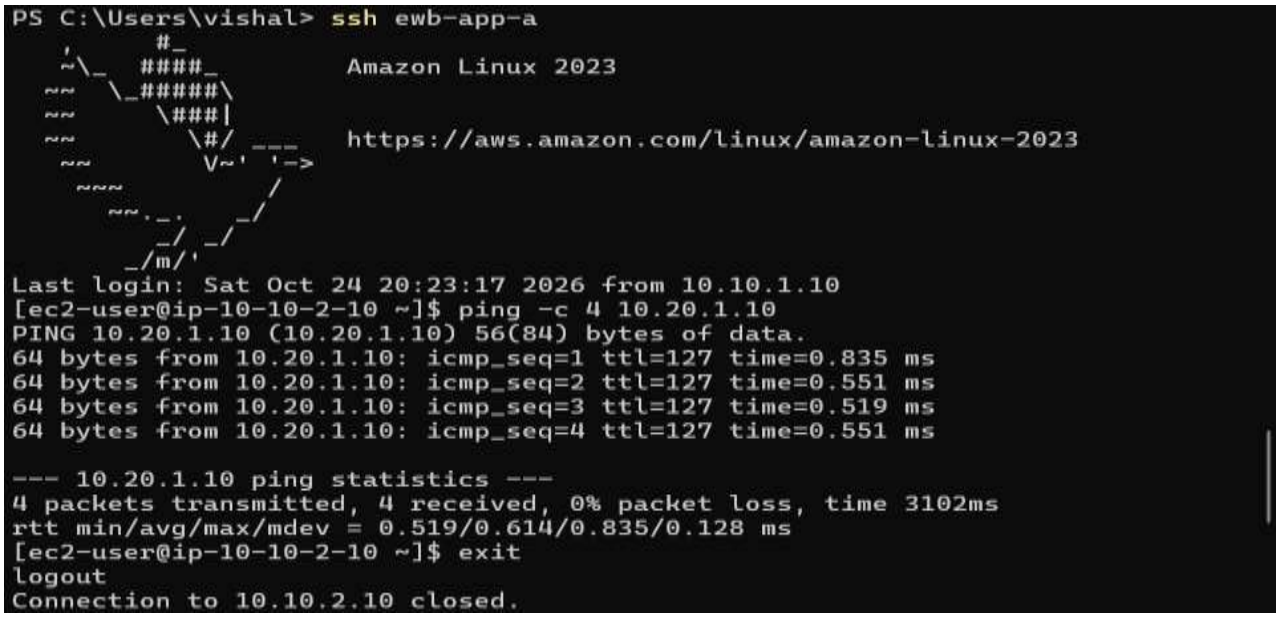
7. Test results

Test	Where/command	Expected	Actual Result
01 Baseline	App-A, before peering: ping -c 4 10.20.1.10	No replies	4 sent, 0 received, 100% loss
02 Forward path	App-A, after routes: ping -c 4 10.20.1.10	4 replies, 0% loss	4 sent, 4 received, 0% loss (avg 0.614 ms)
03 Reverse path	App-B, after routes: ping -c 4 10.10.2.10	4 replies, 0% loss	4 sent, 4 received, 0% loss (avg 0.522 ms)
04 Private SSH	SH Laptop: ssh ewb-app-b	Shell on App-B via bastion	Shell obtained: ec2-user@ip-10-20-1-10

05 Isolation	EC2 instance details: App-A and App-B	No public IPv4	No public IPv4 on either host



Test 01 - App-A (10.10.2.10), before peering: ping to 10.20.1.10 gets no replies (0 received, 100% packet loss)



Test 02 - App-A, after routes: ping to 10.20.1.10 gets 4 replies, 0% packet loss, via the peering connection.

```
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.533 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.528 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.519 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.510 ms

--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3099ms
rtt min/avg/max/mdev = 0.510/0.522/0.533/0.008 ms
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.020 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.027 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.032 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.027 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3133ms
rtt min/avg/max/mdev = 0.020/0.026/0.032/0.004 ms
[ec2-user@ip-10-20-1-10 ~]$
```

1. Test 03 and 04 - shell on App-B (ec2-user@ip-10-20-1-10) reached through the bastion; ping to 10.10.2.10 gets 4 replies, 0% loss.

8. What I learned

Peering is only the link: traffic needs routes on both sides and security rules on the destination. A one-way route is not enough, because replies need the way back. Active status alone does not prove reachability, so each claim needs a ping or SSH test. A bastion with Proxy Jump lets private servers be administered without public IPs or copying keys. Peering is one-to-one and non-transitive and does not share an internet gateway.